IP ADDRESSES:

IPv4 Address Classes:

- **Class A**: 1.0.0.0 to 127.255.255.255
- **Class B**: 128.0.0.0 to 191.255.255.255
- **Class C**: 192.0.0.0 to 223.255.255.255
- **Class D** (Multicast): 224.0.0.0 to 239.255.255.255
- **Class E** (Reserved for future use): 240.0.0.0 to 255.255.255.255

Private IP Address Ranges:

- **Class A**: 10.0.0.0 to 10.255.255.255
- **Class B**: 172.16.0.0 to 172.31.255.255
- **Class C**: 192.168.0.0 to 192.168.255.255

Loopback IP Range:

- **Loopback**: 127.0.0.0 to 127.255.255.255

APIPA (Automatic Private IP Addressing) Range:

- **APIPA**: 169.254.0.0 to 169.254.255.255

Reserved IP Addresses:

- **Network Address**: First address in any subnet (e.g., 192.168.1.0).
- **Broadcast Address**: Last address in any subnet (e.g., 192.168.1.255).

FUNCTIONS OF EACH LAYER IN OSI MODEL:

---

1. Physical Layer:

- Converts **bits** into electrical, optical, or radio signals for transmission.
- **Synchronization** of bit streams between devices.
- Manages **physical connections** (cables, ports, etc.).

---

2. Data Link Layer:

- **Framing**: Divides data into frames for transmission.
- **Error detection and correction**: Using CRC or parity bits.
- **Flow control**: Prevents sender from overwhelming the receiver.
- **MAC addressing**: Adds physical (MAC) addresses for communication.

---

3. Network Layer:

- **Logical addressing**: Assigns IP addresses (IPv4/IPv6).
- **Routing**: Finds the best path for data transmission.
- **Fragmentation**: Breaks down large packets into smaller ones.
- Handles **packet switching**.

---

4. Transport Layer:

- **Segmentation and reassembly**: Divides data into segments, reassembles at destination.
- **Error recovery**: Ensures reliable delivery (TCP).
- **Port addressing**: Distinguishes different applications using port numbers.
- **Flow control**: Manages data rate between devices.

---

5. Session Layer:

- Establishes, manages, and terminates **sessions** between applications.
- **Synchronization**: Inserts checkpoints in long transmissions.
- **Dialog control**: Maintains orderly communication (half-duplex/full-duplex).

---

6. Presentation Layer:

- **Data translation**: Converts data between application and network formats.
- **Encryption and decryption**: Secures data during transmission.
- **Compression**: Reduces data size for faster transmission.

---

7. Application Layer:

- **User interface**: Provides access to network services (browsers, email clients).
- **Protocol management**: Ensures proper use of application protocols (HTTP, FTP).
- **Data formatting**: Handles high-level formatting like JSON, HTML.

ROUTING PROTOCOLS:

---

1. RIP (Routing Information Protocol)

- **Type**: Distance Vector Protocol
- **Key Functions**:
  - Uses **hop count** as the metric (maximum 15 hops).
  - Periodically sends **entire routing tables** to neighbors (every 30 seconds).
  - Implements **split horizon** and **hold-down timers** to avoid routing loops.
- **Strengths**: Simple to configure and use.
- **Weaknesses**:
  - Slow convergence.
  - Not scalable for large networks (limited hop count).

---

2. OSPF (Open Shortest Path First)

- **Type**: Link-State Protocol
- **Key Functions**:
  - Uses **Dijkstra's algorithm** to calculate the shortest path.
  - Divides network into **areas** for scalability (e.g., Area 0 as the backbone).
  - Sends **link-state advertisements (LSAs)** to update only changes in topology.
  - Supports **equal-cost multi-path (ECMP)** for load balancing.
  - Performs **hierarchical routing** with areas.
- **Strengths**:
  - Fast convergence.
  - Scalable for large, complex networks.
  - Supports VLSM and CIDR for efficient IP address usage.
- **Weaknesses**:
  - More complex to configure compared to RIP.

---

3. EIGRP (Enhanced Interior Gateway Routing Protocol)

- **Type**: Advanced Distance Vector Protocol (Hybrid)
- **Key Functions**:
  - Uses **DUAL (Diffusing Update Algorithm)** for fast convergence and loop prevention.
  - Metric based on a combination of **bandwidth**, **delay**, **load**, and **reliability**.
  - Sends **partial updates** (only changes) instead of the entire routing table.
  - Supports **VLSM, CIDR**, and **load balancing** (both equal and unequal cost).

    - Maintains neighbor relationships via **Hello packets**.
- **Strengths**:
    - Fast convergence.
    - Efficient and scalable.
    - Reduced bandwidth usage compared to RIP.
- **Weaknesses**:
    - Proprietary to Cisco (though limited support exists in other vendors).

Comparison Table:

Feature	RIP	OSPF	EIGRP
Type	Distance Vector	Link-State	Hybrid
Metric	Hop Count	Cost (Bandwidth)	Bandwidth & Delay
Convergence Speed	Slow	Fast	Very Fast
Scalability	Low	High	High
Updates	Periodic	Triggered LSAs	Partial/Triggered
Vendor Support	Open Standard	Open Standard	Cisco Proprietary

TIMERS IN ROUTING PROTOCOLS:

1. RIP (Routing Information Protocol)

Timers:

- **Update Timer**: 30 seconds
  - Interval at which RIP routers broadcast their entire routing table.
- **Invalid Timer**: 180 seconds
  - Time after which a route is marked as invalid if no updates are received.
- **Hold-down Timer**: 180 seconds
  - Time during which no changes to a failed route are accepted to avoid loops.
- **Flush Timer**: 240 seconds
  - Time after which the route is completely removed from the table.

Packets:

- **Request Packet**: Used to request routing information from neighbors.
- **Response Packet**: Contains routing table updates sent to neighbors.

---

2. OSPF (Open Shortest Path First)

Timers:

- **Hello Timer**:
  - Default: **10 seconds** (on broadcast and point-to-point links).
  - Default: **30 seconds** (on non-broadcast multi-access (NBMA) links).
  - Determines the interval for sending Hello packets to maintain neighbor relationships.
- **Dead Timer**:
  - Default: **4x the Hello Timer** (40 seconds or 120 seconds).
  - Time after which a neighbor is considered down if no Hello packets are received.
- **LS Refresh Timer**: 30 minutes (1800 seconds)
  - Interval at which LSAs (Link-State Advertisements) are refreshed.
- **LSA Aging Timer**: 60 minutes
  - Time after which LSAs expire if not refreshed.

Packets:

- **Hello Packet**: Establishes and maintains neighbor relationships.
- **Database Description (DBD)**: Summary of LSAs in the router's database.
- **Link-State Request (LSR)**: Requests specific LSAs from neighbors.
- **Link-State Update (LSU)**: Contains updated LSAs sent to neighbors.

- **Link-State Acknowledgment (LSAck)**: Acknowledges receipt of LSAs.

---

3. EIGRP (Enhanced Interior Gateway Routing Protocol)

Timers:

- **Hello Timer**:
  - Default: **5 seconds** (on most links).
  - Default: **60 seconds** (on low-speed links like T1 or below).
  - Determines the interval for sending Hello packets to establish/maintain neighbors.
- **Hold Timer**:
  - Default: **3x the Hello Timer** (15 seconds or 180 seconds).
  - Time after which a neighbor is declared down if no Hello packets are received.
- **Active Timer**: 3 minutes (180 seconds)
  - Maximum time a query remains in the active state before being declared unreachable.

Packets:

- **Hello Packet**: Establishes and maintains neighbor relationships.
- **Update Packet**: Sends routing updates (partial or full) to neighbors.
- **Query Packet**: Queries neighbors for alternate routes during route recalculations.
- **Reply Packet**: Response to a Query, providing routing information.
- **ACK Packet**: Acknowledges the receipt of reliable packets (e.g., Update, Query, Reply).

Summary Table:

Protocol	Timers	Key Packets
RIP	Update: 30s, Invalid: 180s, Hold-down: 180s, Flush: 240s	Request, Response
OSPF	Hello: 10/30s, Dead: 40/120s, LS Refresh: 30m, LSA Aging: 60m	Hello, DBD, LSR, LSU, LSAck
EIGRP	Hello: 5/60s, Hold: 15/180s, Active: 180s	Hello, Update, Query, Reply, ACK

CISCO IOS CONFIGURATION REGISTER:

- 0x2142:
    - Bypasses the NVRAM startup configuration during the boot process.
    - This is typically used for password recovery or when you want to boot the device without applying the saved configuration.
- 0x2100:
    - Boots the device into ROMMON mode (ROM Monitor mode).
    - This is used for troubleshooting, firmware recovery, or manual boot configuration.
- 0x2101:
    - Boots the device into Mini IOS (also known as RXBOOT mode or Bootstrap mode).
    - This mode provides a minimal IOS image, often used for diagnostics or basic operations.
- 0x2102 to 0x210F:
    - Boots the device normally using the NVRAM startup configuration.
    - 0x2102 is the default configuration register value for most devices, ensuring a normal boot sequence.

COMMON PROTOCOLS YOU MUST KNOW:

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1. TCP/IP (Transmission Control Protocol / Internet Protocol)

- **TCP**: Connection-oriented protocol ensuring reliable, ordered data transmission.
- **IP**: Responsible for addressing and routing packets across networks (IPv4 and IPv6).

2. ARP (Address Resolution Protocol)

- Resolves **IP addresses** to **MAC addresses** on a local network.
- Essential for devices to communicate within the same network.

3. DNS (Domain Name System)

- Converts human-readable **domain names** (e.g., www.example.com) into **IP addresses**.
- Crucial for web browsing and most internet services.

4. HTTP/HTTPS (HyperText Transfer Protocol / Secure HTTP)

- **HTTP**: Used for transferring web pages and resources.
- **HTTPS**: A secure version of HTTP that uses **SSL/TLS** to encrypt communication.

5. DHCP (Dynamic Host Configuration Protocol)

- Automatically assigns **IP addresses** and other network configuration details to devices on a network.

6. FTP (File Transfer Protocol)

- Protocol for transferring files between computers over a network.
- Includes both active and passive modes for connection establishment.

7. SMTP (Simple Mail Transfer Protocol)

- Used for sending **email** messages between servers.
- Typically works alongside **IMAP** or **POP3** for receiving emails.

8. POP3/IMAP (Post Office Protocol / Internet Message Access Protocol)

- **POP3**: Used to download emails from the server to the client.
- **IMAP**: Provides more advanced features, allowing email synchronization across multiple devices.

9. ICMP (Internet Control Message Protocol)

- Primarily used for **error reporting** and diagnostics (e.g., **ping** command).
- Helps with network troubleshooting and ensuring proper routing.

10. SNMP (Simple Network Management Protocol)

- Used for monitoring and managing network devices like routers, switches, and servers.
- Enables administrators to gather data about network performance and health.

11. Telnet/SSH (Secure Shell)

- **Telnet**: An older, unencrypted protocol for accessing remote systems.
- **SSH**: A secure, encrypted protocol for remote access and management of devices.

12. BGP (Border Gateway Protocol)

- A **routing protocol** used to exchange routing information between different **autonomous systems** on the internet.
- It's the protocol that enables routing between large networks and the internet.

13. OSPF (Open Shortest Path First)

- **Link-state routing protocol** used within an **autonomous system**.
- Helps routers find the best path to route packets based on a **cost** metric.

14. RIP (Routing Information Protocol)

- **Distance-vector routing protocol** used for small to medium-sized networks.
- Operates by exchanging entire routing tables periodically.

15. EIGRP (Enhanced Interior Gateway Routing Protocol)

- A hybrid routing protocol (combining distance vector and link-state features).
- Known for fast convergence and efficient bandwidth usage.

16. VPN (Virtual Private Network)

- Allows secure communication over the internet by creating a **tunnel** between the client and server.
- Common protocols: **PPTP**, **L2TP**, **IPsec**, **SSL/TLS**.

17. NTP (Network Time Protocol)

- Synchronizes the clocks of computers over a network to maintain accurate time across devices.

18. SSL/TLS (Secure Sockets Layer / Transport Layer Security)

- **SSL** is the predecessor to **TLS**, both of which are used for encrypting data during transmission.
- Commonly used in **HTTPS** for securing web traffic.

BASIC COMMANDS OF ROUTING PROTOCOLS:

RIP (Routing Information Protocol) Commands:

1. **Enable RIP routing**:
   `router rip`

2. **Define RIP version**:
   `version <1 | 2>` (Version 1 or 2)

3. **Enable RIP on specific networks**:
   `network <network_address>`

4. **Disable RIP on specific interfaces**:
   `passive-interface <interface_name>`

5. **Set default network (default route advertisement)**:
   `default-information originate`

6. **Show RIP routing table**:
   `show ip rip database`

7. **Show RIP routes**:
   `show ip route rip`

OSPF (Open Shortest Path First) Commands:

1. **Enable OSPF routing**:
   `router ospf <process_id>`

2. **Define OSPF networks**:
   `network <network_address> <wildcard_mask> area <area_id>`

3. **Set router ID for OSPF**:
   `router-id <router_id>`

4. **Enable OSPF on specific interface**:
   ```
   ip ospf <process_id> area <area_id>
   ```

5. **View OSPF neighbors**:
   ```
   show ip ospf neighbor
   ```

6. **View OSPF routing table**:
   ```
   show ip ospf route
   ```

7. **Show OSPF database**:
   ```
   show ip ospf database
   ```

EIGRP (Enhanced Interior Gateway Routing Protocol) Commands:

1. **Enable EIGRP routing**:
   ```
   router eigrp <AS_number>
   ```

2. **Define EIGRP networks**:
   ```
   network <network_address> <wildcard_mask>
   ```

3. **Set EIGRP metrics**:
   ```
   metric weights 0 <bandwidth> <delay> <reliability> <load>
   ```

4. **Enable EIGRP on a specific interface**:
   ```
   ip hello-interval eigrp <AS_number> <seconds>
   ip hold-time eigrp <AS_number> <seconds>
   ```

5. **View EIGRP neighbors**:
   ```
   show ip eigrp neighbors
   ```

6. **Show EIGRP routing table**:
   ```
   show ip route eigrp
   ```

7. **Show EIGRP topology table**:
   ```
   show ip eigrp topology
   ```

Static Routing Commands:

1. **Add a static route**:
   `ip route <destination_network> <subnet_mask> <next_hop_address>`

2. **Add a static route to a specific interface**:
   `ip route <destination_network> <subnet_mask> <exit_interface>`

3. **View the routing table**:
   `show ip route`

4. **Delete a static route**:
   `no ip route <destination_network> <subnet_mask> <next_hop_address>`

NAT (Network Address Translation) Commands:

Static NAT:

1. **Configure Static NAT**:
   `ip nat inside source static <local_ip> <global_ip>`

2. **Show NAT translation table**:
   `show ip nat translations`

3. **Configure Inside and Outside interfaces**:
   `ip nat inside` (for inside interface)
   `ip nat outside` (for outside interface)

Dynamic NAT:

1. **Define Dynamic NAT Pool**:
   `ip nat pool <pool_name> <start_ip> <end_ip> netmask <subnet_mask>`

2. **Map inside source to dynamic pool**:
   `ip nat inside source list <access_list_number> pool <pool_name>`

3. **Create Access Control List (ACL)**:
   ```
   access-list <access_list_number> permit <source_network>
   <wildcard_mask>
   ```
4. **Show NAT configuration**:
   ```
   show running-config | include nat
   ```
5. **Show Dynamic NAT translations**:
   ```
   show ip nat translations
   ```

IPv4 HEADER:

1. **Version** (4 bits) – Specifies the IP version (IPv4).
2. **IHL (Internet Header Length)** (4 bits) – Specifies the length of the header.
3. **Type of Service (TOS)** (8 bits) – Used to specify quality of service features.
4. **Total Length** (16 bits) – The total length of the IP packet (header + data).
5. **Identification** (16 bits) – Unique identifier for the packet.
6. **Flags** (3 bits) – Control flags for fragmentation.
7. **Fragment Offset** (13 bits) – Specifies the position of the fragment in the original packet.
8. **Time to Live (TTL)** (8 bits) – Specifies the maximum hops the packet can make.
9. **Protocol** (8 bits) – Defines the protocol used in the data section (e.g., TCP, UDP).
10. **Header Checksum** (16 bits) – Error-checking for the header.
11. **Source IP Address** (32 bits) – The sender's IP address.
12. **Destination IP Address** (32 bits) – The receiver's IP address.
13. **Options** (variable) – Optional fields (not always present).
14. **Padding** (variable) – Padding to ensure the header is a multiple of 32 bits.

Total Size: 20 to 60 bytes (20 bytes without options).

ETHERNET HEADER:

- **Destination MAC Address (6 bytes)**: Specifies the recipient's MAC address.
- **Source MAC Address (6 bytes)**: Specifies the sender's MAC address.
- **EtherType / Length (2 bytes)**: Indicates the protocol (e.g., IPv4, ARP) or, in some cases, the payload length (if the value is less than or equal to 1500).
- **Payload (46 to 1500 bytes)**: The actual data being transmitted. The minimum size of the payload is 46 bytes to ensure the frame is at least 64 bytes in total.
- **Frame Check Sequence (FCS) (4 bytes)**: Used for error detection via CRC (Cyclic Redundancy Check).

Total Size: 14 bytes (Ethernet header, excluding the FCS).

IEEE STANDARDS FOR TECHNOLOGIES:

Ethernet and Related Standards:

- **Ethernet (Wired)** - IEEE 802.3
- **Ethernet MAC** - IEEE 802.3
- **Gigabit Ethernet** - IEEE 802.3z
- **10 Gigabit Ethernet** - IEEE 802.3ae
- **Fast Ethernet** - IEEE 802.3u

Wireless Standards:

- **Wi-Fi (Wireless)** - IEEE 802.11
- **Wireless LAN MAC** - IEEE 802.11
- **Wireless Security (WPA2)** - IEEE 802.11i

Other Network Topologies:

- **Token Ring** - IEEE 802.5
- **Token Ring MAC** - IEEE 802.5
- **Bluetooth** - IEEE 802.15.1
- **Bluetooth MAC** - IEEE 802.15.1

Networking Protocols and Management:

- **Bridges** - IEEE 802.1D
- **Spanning Tree Protocol (STP)** - IEEE 802.1D
- **VLANs (Virtual LANs)** - IEEE 802.1Q

CABLES AND THEIR SPEEDS:

Ethernet (Wired):

- **Cat 5 (Twisted Pair Cable)**: Speed: Up to **100 Mbps** (Fast Ethernet)
- **Cat 5e (Enhanced Twisted Pair Cable)**: Speed: Up to **1 Gbps** (Gigabit Ethernet)
- **Cat 6 (Twisted Pair Cable)**: Speed: Up to **10 Gbps** (10 Gigabit Ethernet) at shorter distances (~55 meters)

Fiber Optic Cables:

- **Single-Mode Fiber (SMF)**: Speed: Up to **100 Gbps** and beyond (Long-distance transmission)
- **Multi-Mode Fiber (MMF)**: Speed: Up to **100 Gbps** (Shorter distances, typically up to 2 km)

Wi-Fi (Wireless) Standards:

- **Wi-Fi 4 (802.11n)**: Speed: Up to **600 Mbps** (with 40 MHz channels)
- **Wi-Fi 5 (802.11ac)**: Speed: Up to **3.5 Gbps** (with multiple channels)
- **Wi-Fi 6 (802.11ax)**: Speed: Up to **9.6 Gbps** (with optimal conditions and multiple channels)

Other Networking Technologies:

- **Token Ring (Twisted Pair Cable)**: Speed: Up to **16 Mbps**
- **Ethernet over Coaxial (10BASE2, 10BASE5)**: Speed: Up to **10 Mbps**
- **DSL (Digital Subscriber Line)**: Speed: Up to **100 Mbps** (depending on the type: ADSL, VDSL)

Wireless Network Standards:

- **Bluetooth 4.0 (Low Energy)**: Speed: Up to **25 Mbps**
- **Bluetooth 5.0**: Speed: Up to **50 Mbps**
- **Zigbee**: Speed: Up to **250 Kbps** (for IoT applications)

Command: A#show int g0/0

- **Interface is up, Line Protocol is up**:
  The interface is working fine, and both the physical connection (Layer 1) and the data connection (Layer 2) are active. The connection is successful.
- **Interface is down, Line Protocol is down**:
  This could mean:
  - The interface has been manually turned off using the `shutdown` command.
  - There's no physical connection (Layer 1), or the data link (Layer 2) isn't working.
- **Interface is up, Line Protocol is down**:
  This means there's a problem with the data link (Layer 2). Possible causes could be:
  - Clocking mismatch
  - Mismatched encapsulation
  - Keep-alive failure
- **Interface is administratively down, Line Protocol is down**:
  The interface has been manually disabled using the `shutdown` command. To fix this, you need to use the `no shutdown` command.
- **Note**: There's never a situation where the interface is down, but the line protocol is up.

PORTS:

- **0 to 1023**: **Well-known ports** (used by widely known services like HTTP, FTP, DNS)
- **1024 to 49151**: **Registered ports** (used by applications and services not in the well-known list)
- **49152 to 65535**: **Dynamic/Private ports** (used for ephemeral, temporary connections)

Class A: 127 networks and 16,777,216 nodes per network

Class B: 16,384 networks and 65,534 nodes per network

Class C: 2,097,152 networks and 254 nodes per network